THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 10: **Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A lens produces a bright spot on paper when held in sunlight. The lens is: 1

(a) concave **(b)** convex
(c) cylindrical glass **(d)** plane

2. A ray incident on a mirror gets reflected such that the reflected ray is perpendicular to the incident ray. The angle of incidence is: 1

(a) 60 degrees **(b)** 30 degrees **(c)** 90 degrees **(d)** 45 degrees

3. Which of the following mirrors can form both real and virtual images? 1

(a) concave **(b)** plane **(c)** cylindrical **(d)** convex

4. A beam of parallel rays passes through a concave lens. The rays after refraction: 1

(a) stop **(b)** stay parallel **(c)** diverge **(d)** converge

5. The image of a clock in a plane mirror shows 3:45. The actual time is: 1

(a) 4:15 **(b)** 9:15 **(c)** 3:45 **(d)** 8:15

6. The focal length of a plane mirror is: 1

(a) 1 m **(b)** zero **(c)** infinity **(d)** -1 m

7. A student finds that a lens always gives a smaller erect image. The lens is: 1

(a) concave **(b)** plane **(c)** cylindrical **(d)** convex

8. Which letter looks the same when seen in a plane mirror (apart from left-right reversal)? 1

(a) A **(b)** F **(c)** B **(d)** R

9. A dentist needs a mirror that gives an enlarged image. She uses: 1

(a) concave mirror **(b)** a prism **(c)** convex mirror **(d)** plane mirror

10. The human eye lens is a: 1

(a) concave lens **(b)** plane mirror **(c)** convex lens **(d)** prism

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do the surfaces of mirrors have a polished metal or silver coating? 2

12. An object is 20 cm from a plane mirror. How far is the image from the object if the object moves 5 cm away? 2

13. What is meant by the optical centre of a lens? 2

14. What type of lens is used in a camera, a projector and a pair of spectacles for myopia? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare convex and concave lenses based on shape, action on rays and uses. 3

16. State three differences between a plane mirror and a spherical mirror. 3

17. Explain why the sky looks dark at night but a mirror still shows our image in the room light. 3

OR

Explain the use of a convex lens as a magnifying glass. 3

18. Calculate the focal length of a concave mirror whose radius of curvature is 24 cm and tell where a parallel beam will converge. 3

OR

Why do some shops use convex mirrors? Give two reasons. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss the uses of reflecting and refracting devices: plane, concave, convex mirrors; convex and concave lenses, with a daily-life example each. 5

OR

Explain how sunlight can be used to light a piece of paper with a convex lens, in terms of rays and energy. 5

20. Explain the working and construction of a periscope and kaleidoscope. Mention the uses and principle. 5

OR

Discuss why images in mirrors appear laterally inverted and not inverted vertically. Show with the letter F and the number 3. 5

END OF QUESTION PAPER — Total: 40 Marks

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